



Inspiring Excellence

CSE340

Summer 2025

Assignment 2

Name : Abdullah Al Fahad

ID : 21201240

Section : 8

Faculty : PBK

Assignment 2

1) By looking at the opcode field we understand what operation / type of Instructions the CPU will perform.

In the opcode field a set of bits specify for the responsible function. Such R, I type Instructions have different type of opcode and by looking at op-code we can ensure the instructions type.

Moreover, OP-code includes ~~source/destination register location~~ and type of operation such add, sub etc.

2) In op-code ~~field~~ field by looking at funct3 it indicates the size of data being loaded. for LD instruction D means double word which is 64 bit - this information stored in funct3 in opcode.

9) The funct3 field in load instruction specifies the size of the data and signed and unsigned value:

Such, 010 $\rightarrow$ LW (signed) $\rightarrow$ 32 bit

011 $\rightarrow$ LD $\rightarrow$ 64 bit

So, without funct3 function the CPU

Day : _____

Time : _____ Date : / /

wouldn't know ~~how~~ how many bits to load.

5) In S-type instruction the immediate value is divided into two portion

$\text{imm}[11:5] \mid \text{rs2} \mid \text{rs1} \mid \text{imm3} \mid \text{imm}[4:0] \mid \text{opcode}$

because to ~~keep~~ maintain similarity in structure, hardware simplicity, and efficient use of space.

3) a)

A, B = Double word
 $\Rightarrow 64/8 \Rightarrow 8$

$$\alpha = [(A[B[3]] + B[24] - 12) * 28] / 256$$

A = X20

B = X21

$\alpha = X22$

LD X5, 24(X21)

// B[3]

SHL X6, X5, 3

// $X5 \times 2^3$

Add X7, X20, X6

// $A[B[3]]$
// (Base + offset)

ld ~~X8~~, 0(X7)

// $X8 \Rightarrow A[B[3]]$

LD X5, 192(X21)

// $X5 \Rightarrow B[12]$

~~Add ~~X8~~ X5, X5, -12~~ // $X5 \Rightarrow B[12] - 12$

Day: _____

Time: _____ Date: / /

Add X_9, X_8, X_5 // $A[B[3]] + B[24]$

Addi $X_9, X_9, -12$ // $A[B[3]] + B[24] - 12$

Slli $X_4, X_9, 5$ // ~~Let~~ $X_4 = (A[B[3]] + B[24] - 12)$

$X_4 = \cancel{X_4} \times 32$

Slli $X_6, X_9, 2$ // $X_6 = \cancel{X_4} \times 4$

Sub X_9, X_4, X_6 // $X_9 = \left[(A[B[3]] + B[24] - 12) \times 28 \right]$

SRli

~~Srli $X_9, X_9, 8$ // $\cancel{X_9} \times 8$~~

Srli $X_{22}, X_9, 8$ // $X_{22} = \left[(A[B[3]] + B[24] - 12) \times 28 \right] / 256$

Day : _____

Time : _____ Date : / /

(6)

R-typeAdd X_2, X_{20}, X_6

func7	rs2	rs1	func3	rd	opcode
XXXXXX	00110	10100	XXX	00111	0110011

I-typeSll $X_4, X_9, 5$

func6 ↓ Immediate	rs1	func3	rd	opcode
00000000101	01001	XXX	00100	XXXXXXXX

Day: _____

Time: _____ Date: / /

S-type

SD, X21, 10(X22)

Imm[11:5]	rs2	rs1	funct3 funct3	Imm[9:0]	opcode
0000000	10101	10110	xxx	01010	xxxxxxx